

Physical Chemistry

Topic- Atomic Structure

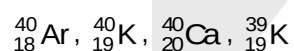
Work Sheet - 04

1. Complete the following table :

Particle	Mass No.	Atomic No.	Protons	Neutrons	Electrons
Nitrogen atom	—	—	—	7	7
Calcium ion	—	20	—	20	—
$^{80}_{35}\text{Br}$	80	—	—	—	—
$^{37}\text{Cl}^-$	—	17	—	—	—
Na^+	23	—	—	—	—

2. The number of electrons, protons and neutrons in a species are equal to 18, 16 and 16 respectively. Assign the proper symbol to the species.

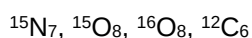
3. Identify the isobars?



4. Identify the isoelectronic species with N^{3-} ?



5. Identify the isotones of $^{14}\text{C}_6$?



Complete the following table :

Q.No.	Frequency	Wavelength	Energy of one photon	Wave number (In m^{-1})
1.	(in Hz)	400 nm	(In eV)	
2.	3×10^{14} Hz	(In m)	(In kJ)	
3.	(in sec^{-1})	(In $\text{\AA}$)	2 eV	

4. Find number of photons emitted per second by 300 watt bulb emitting light of wavelength 660 nm.

5. Find number of photons emitted per minute by 40 watt bulb with 50% efficiency if wavelength of emitted is 3300 $\text{\AA}$.

Q. Complete the following table :

Q.No.	Mass of particle	Velocity of particle	de-Broglie wavelength
1.	0.1 kg	10 m/s	
2.		10^6 m/s	13. 2 Å
3.	3×10^{-31} g		220 nm

4. Find de-Broglie wavelength when an electron has a kinetic energy of 1.5 eV.

5. Find de-Broglie wavelength when an electron having initial kinetic energy of 1.2×10^{-19} kJ, is accelerated through a potential difference 0.75 V.

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Work Sheet - 07

Complete the following table:

Q. No.	ν_0 (threshold)	ν (incident)	Maximum Kinetic energy
1		1.1×10^{15} Hz	3.313×10^{-19} J
2	4×10^{10} Hz	6×10^9 Hz	
3	7×10^{14} Hz		1.988×10^{-19} J

Complete the following table:

Q. No.	Work function	λ (incident)	Maximum Kinetic energy
4	2.5 eV		1.5 eV
5	2.31×10^5 J	300 nm	